Week 5: Volumes, determinants, and the adjugate

Linear Algebra I
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Fall 2026

These notes were drafted with the help of ChatGPT from my handwritten notes and the discussion used to prepare the recitation.

The determinant connects geometric volume with matrix operations and linear systems. We begin by recalling this geometric meaning, then develop the adjugate and Cramer's rule before applying these ideas to block determinants.

1. Volumes and Gram determinants

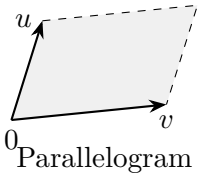
For $u, v \in \mathbb{R}^2$, the area of their parallelogram is $|\det(u\ v)|$. For $u, v \in \mathbb{R}^3$, the corresponding area is $\|u \times v\|$, where

$$u \times v = (u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1)^T.$$

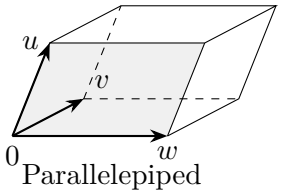
For three vectors $u, v, w \in \mathbb{R}^3$, the parallelepiped and the tetrahedron with vertices $0, u, v, w$ have volumes

$$|\det(u\ v\ w)|, \qquad \frac{1}{6}|\det(u\ v\ w)|,$$

respectively. The determinant is signed; ordinary area and volume are nonnegative.



Parallelogram



Parallelepiped

The same question makes sense for k vectors in $\mathbb{R}^n$, even when their coordinate matrix is not square. Let $1 \le k \le n$ and set

$$A = (u_1\ \cdots\ u_k) \in M_{n,k}(\mathbb{R}), \qquad \mathcal{P}(A) = \left\{ \sum_{j=1}^{k} t_j u_j : 0 \le t_j \le 1 \right\}.$$

The matrix $G = A^TA$, whose (i, j) entry is $u_i \cdot u_j$, is the *Gram matrix*. The k-dimensional volume is

$$\mathrm{Vol}_k\, \mathcal{P}(A) = \sqrt{\det(A^TA)}.$$

The simplex with vertices $0, u_1, \ldots, u_k$ has $1/k!$ of this volume. The factor follows by repeated use of the pyramid formula: base volume times height, divided by the dimension. In three dimensions this recovers the factor $1/6$ above.

Proof of the Gram determinant formula

Suppose first that $u_1, \dots, u_k$ are linearly independent. Gram–Schmidt gives a thin QR factorization

$$A = QR, \qquad Q^T Q = I_k,$$

where $Q \in M_{n,k}(\mathbb{R})$ has orthonormal columns and $R \in M_k(\mathbb{R})$ is upper triangular with positive diagonal entries. In the orthonormal coordinates provided by Q, the columns of A have coordinate matrix R.

More explicitly, Gram–Schmidt constructs

$$w_j = u_j - \sum_{i<j} (q_i \cdot u_j) q_i, \qquad q_j = \frac{w_j}{\|w_j\|}, \qquad r_{jj} = \|w_j\|.$$

The number r_{jj} is the height of u_j above the span of the preceding vectors. Repeated base-times-height therefore gives

$$\operatorname{Vol}_k \mathcal{P}(A) = \prod_{j=1}^{k} r_{jj} = \det R.$$

On the other hand,

$$A^T A = R^T Q^T Q R = R^T R, \qquad \det(A^T A) = (\det R)^2.$$

Taking the nonnegative square root proves the formula.

If the columns are linearly dependent, the k-dimensional volume is zero. There is a nonzero $z \in \mathbb{R}^k$ with $Az = 0$, so $A^T A z = 0$ as well. Thus the Gram matrix is singular, and the same formula holds. When $k = n$, it reduces to

$$\sqrt{\det(A^T A)} = \sqrt{(\det A)^2} = |\det A|.$$

2. Basic determinant identities

From now on, unless otherwise stated, $A, B \in M_n(\mathbb{R})$ are square matrices. We will repeatedly use

$$\det(cA) = c^n \det A, \qquad \det(AB) = \det A \det B.$$

The first identity scales all n columns; the second says that volume factors multiply under composition. They imply

$$\det(A^m) = (\det A)^m \quad (m \geq 1), \qquad \det(A^{-1}) = (\det A)^{-1} \quad (A \text{ invertible}).$$

For every invertible P, we also have

$$\det(P^{-1} A P) = \det A, \qquad \det(A^T) = \det A.$$

The first follows by cancellation of $\det P$; the second exchanges the row and column descriptions of the same determinant. Recall also that swapping two rows changes the sign, scaling one row scales the determinant, and adding a multiple of another row leaves it unchanged. The corresponding column statements hold as well.

In particular, a triangular matrix has determinant equal to the product of its diagonal entries, and A is invertible exactly when $\det A \neq 0$. Linearity in one column should not be confused with additivity in the whole matrix: in general, $\det(A + B) \neq \det A + \det B$.

3. Cofactors and the adjugate

For $n \geq 2$, let M_{ij} be the determinant of the submatrix obtained by deleting row i and column j of A, and put $C_{ij} = (-1)^{i+j} M_{ij}$. The *adjugate* is the transpose of the cofactor matrix:

$$A^{\vee} = (C_{ji})_{i,j}.$$

The notation C_{ij} here is the lecture notes' A_{ij}. It is a scalar, while $A^{\vee}$ is a matrix.

The cofactor expansion and the repeated-row property give

$$\sum_{k=1}^{n} a_{ik} C_{jk} = \delta_{ij} \det A.$$

For $i = j$, this is expansion along row i. For $i \neq j$, replace row j by row i. Its cofactors along row j are unchanged, but its determinant is zero because two rows coincide. Consequently,

$$(AA^{\vee})_{ij} = \sum_{k} a_{ik} C_{jk} = \delta_{ij} \det A.$$

The corresponding column argument gives the other order. Hence

$$AA^{\vee} = A^{\vee} A = (\det A) I_n.$$

This identity holds even when A is singular. If $\det A \neq 0$, division gives

$$A^{-1} = \frac{1}{\det A} A^{\vee}.$$

Example

Consider

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 1 & 1 & 2 \\ 2 & 3 & 4 \end{pmatrix},$$

the signed minors are

$$\begin{array}{lll} C_{11} = 4 - 6 = -2, & C_{12} = -(4 - 4) = 0, & C_{13} = 3 - 2 = 1, \\ C_{21} = -(12 - 3) = -9, & C_{22} = 4 - 2 = 2, & C_{23} = -(3 - 6) = 3, \\ C_{31} = 6 - 1 = 5, & C_{32} = -(2 - 1) = -1, & C_{33} = 1 - 3 = -2. \end{array}$$

Transposing the cofactor matrix gives

$$A^{\vee} = \begin{pmatrix} -2 & -9 & 5 \\ 0 & 2 & -1 \\ 1 & 3 & -2 \end{pmatrix}.$$

Expansion along the first row yields $\det A = -2 + 0 + 1 = -1$, so $A^{-1} = -A^{\vee}$. As a check, direct multiplication gives $AA^{\vee} = -I_3$.

4. Further identities for the adjugate

Throughout this section and the next, assume $n \geq 2$.

The determinant of the adjugate

For every $A \in M_n(\mathbb{R})$,

$$\det(A^\vee) = (\det A)^{n-1}.$$

If A is invertible, take determinants in $AA^\vee = (\det A)I_n$:

$$\det A \det(A^\vee) = (\det A)^n.$$

Dividing by $\det A \neq 0$ proves the claim in this case.

If A is singular and $A \neq 0$, suppose that $A^\vee$ were invertible. Then $AA^\vee = 0$ would imply $A = 0$, a contradiction. Thus $\det(A^\vee) = 0$. If $A = 0$, every $(n-1)$-minor is zero because $n \geq 2$, so $A^\vee = 0$ directly. This proves the identity for singular matrices as well.

Transpose and scalar multiplication

The cofactor of A^T at (i, j) equals the cofactor of A at (j, i). Each minor has order $n-1$, so scaling A by c scales each cofactor by c^{n-1}. Therefore

$$(A^T)^\vee = (A^\vee)^T, \qquad (cA)^\vee = c^{n-1}A^\vee.$$

The scalar identity holds for all $c \in \mathbb{R}$, including $c = 0$.

A product reverses the order

For any two matrices $A, B \in M_n(\mathbb{R})$,

$$(AB)^\vee = B^\vee A^\vee.$$

For invertible A, B, this follows from the inverse formula:

$$\begin{aligned}(AB)^\vee &= \det(AB)(AB)^{-1} \\ &= \det A \det B \, B^{-1}A^{-1} = B^\vee A^\vee.\end{aligned}$$

To include singular matrices, put $A_t = A + tI_n$ and $B_t = B + tI_n$. Each is invertible except at finitely many real t, because its determinant is a monic polynomial of degree n. Thus the identity holds for infinitely many t after substituting A_t, B_t. Every matrix entry on both sides is a polynomial in t, so these polynomials agree identically. Setting $t = 0$ proves the formula for the original A, B.

This polynomial argument is useful whenever a cofactor identity is first obtained for invertible matrices. In particular, invertibility is needed to divide by a determinant, but the resulting polynomial identity may remain valid when that determinant is zero.

5. The rank of the adjugate

For $n \geq 2$,

$$\operatorname{rank}(A^\vee) = \begin{cases} n, & \operatorname{rank} A = n, \\ 1, & \operatorname{rank} A = n-1, \\ 0, & \operatorname{rank} A \leq n-2. \end{cases}$$

If $\operatorname{rank} A = n$, then $A^\vee = (\det A)A^{-1}$ is invertible, so its rank is n. If $\operatorname{rank} A \leq n-2$, every $(n-1)$-minor vanishes, hence every cofactor vanishes and $A^\vee = 0$.

The case $\operatorname{rank} A = n-1$ requires both a lower and an upper bound. First choose $n-1$ linearly independent columns of A, forming an $n \times (n-1)$ matrix B. Its row rank is also $n-1$, so we can choose $n-1$ independent rows of B. They form an invertible square submatrix of order $n-1$, whose determinant is nonzero.

In the original $n \times n$ matrix, choosing $n-1$ rows and $n-1$ columns deletes exactly one row i and one column j. The resulting nonzero minor is M_{ij}, so $C_{ij} \neq 0$. Therefore $A^\vee \neq 0$ and $\operatorname{rank}(A^\vee) \geq 1$.

On the other hand, $\det A = 0$ and $AA^\vee = 0$. Writing $A^\vee = (v_1 \ \cdots \ v_n)$ gives $Av_j = 0$ for every j. All columns lie in

$$\ker A, \qquad \dim \ker A = n - \operatorname{rank} A = 1.$$

Consequently $\operatorname{rank}(A^\vee) \leq 1$, proving equality. For example,

$$A = \operatorname{diag}(1,1,0) \quad \Longrightarrow \quad A^\vee = \operatorname{diag}(0,0,1).$$

Thus singularity of A does not force its adjugate to be zero.

Taking the adjugate twice

The preceding rank result also allows us to compute the adjugate of the adjugate:

$$(A^\vee)^\vee = \begin{cases} A, & n = 2, \\ (\det A)^{n-2} A, & n > 2. \end{cases}$$

For invertible A, use the determinant and inverse of $A^\vee$:

$$(A^\vee)^\vee = \det(A^\vee)(A^\vee)^{-1} = (\det A)^{n-1}(\det A)^{-1}A.$$

For $n = 2$, direct calculation gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^\vee = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and taking the adjugate again returns A, even when A is singular. For $n > 2$ and singular A, we have $\operatorname{rank}(A^\vee) \leq 1 < n-1$. All $(n-1)$-minors of $A^\vee$ therefore vanish, so $(A^\vee)^\vee = 0$. This equals $(\det A)^{n-2}A$ as required.

6. Cramer's rule and volume scaling

The inverse formula expresses the solution of a linear system through cofactors. Cramer's rule packages each coordinate into one determinant.

Let $A = (a_1 \ \cdots \ a_n)$ satisfy $\det A \neq 0$, and let $Ax = b$. Denote by $A_i(b)$ the matrix obtained by replacing column i of A by b. Since $b = \sum_j x_j a_j$, multilinearity in column i gives

$$\det A_i(b) = \sum_{j=1}^{n} x_j \det(a_1, \ldots, a_{i-1}, a_j, a_{i+1}, \ldots, a_n).$$

For $j \neq i$, two columns coincide and the term vanishes. Hence

$$\det A_i(b) = x_i \det A, \qquad x_i = \frac{\det A_i(b)}{\det A}.$$

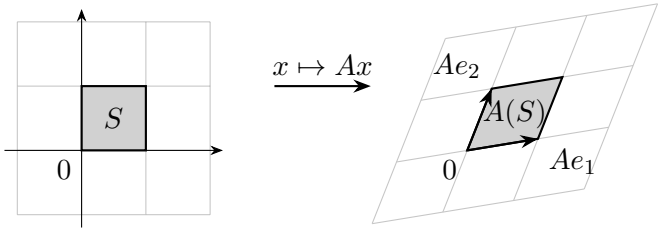
In geometric terms, the coordinate x_i is a ratio of signed volumes. In $\mathbb{R}^3$, the first coordinate has the particularly concrete form

$$x_1 = \frac{b \cdot (a_2 \times a_3)}{a_1 \cdot (a_2 \times a_3)}.$$

Taking the dot product with $a_2 \times a_3$ eliminates the other two coordinates. The adjugate collects these coordinate-extraction operations into a single matrix.

The determinant as an area or volume factor

For $A \in M_n(\mathbb{R})$, the map $T(x) = Ax$ sends e_j to column a_j. Thus it sends the unit cube to the parallelepiped spanned by the columns of A.



Unit square Area $|\det A|$

More generally, for an n-dimensional parallelepiped P,

$$\mathrm{Vol}_n(A(P)) = |\det A| \ \mathrm{Vol}_n(P).$$

Indeed, if P has edge matrix B, its image has edge matrix AB, and $|\det(AB)| = |\det A|\,|\det B|$. A positive determinant preserves orientation, a negative one reverses it, and a zero determinant collapses dimension. When $\det A = 0$, Cramer's ratios cannot be used; the original system must be examined for consistency and free variables.

7. Block elimination and determinants

Exercise 3.3, No. 7

The identities above turn matrix factorizations into determinant formulas. We now apply them to a block version of Gaussian elimination.

Problem. Let $A, B, C, D \in M_n(\mathbb{R})$ and suppose A is invertible. Find X, Y such that

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & 0 \\ X & I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & Y \end{pmatrix}.$$

Deduce a formula for the determinant. If, in addition, $AC = CA$, prove that this determinant equals $\det(AD - CB)$.

Solution. Multiplying the two block matrices gives

$$\begin{pmatrix} I & 0 \\ X & I \end{pmatrix} \begin{pmatrix} A & B \\ 0 & Y \end{pmatrix} = \begin{pmatrix} A & B \\ XA & XB + Y \end{pmatrix}.$$

Matching the lower blocks requires $XA = C$ and $XB + Y = D$. Therefore

$$X = CA^{-1}, \qquad Y = D - CA^{-1}B.$$

The first factor is block lower triangular with identity diagonal blocks, so its determinant is 1. The second is block upper triangular. Consequently,

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \, \det(D - CA^{-1}B).$$

Here we used the block triangular determinant formula: the determinant is the product of the determinants of its square diagonal blocks. Equivalently, the elimination can be written as

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}.$$

The eliminating factor has determinant 1, just as in ordinary row elimination.

If $AC = CA$, then $ACA^{-1} = C$. Multiply A into the remaining factor on the left:

$$\begin{aligned} \det A \det(D - CA^{-1}B) &= \det\big(A(D - CA^{-1}B)\big) \\ &= \det(AD - ACA^{-1}B) \\ &= \det(AD - CB). \end{aligned}$$

The order of the factors is essential. The expression $D - CA^{-1}B$ comes from the actual block multiplication; its factors cannot be rearranged arbitrarily. The simpler expression $\det(AD - CB)$ is justified here by the commuting condition $AC = CA$.

8. Two eliminations of the same block matrix

Exercise 3.3, No. 8: equivalent plus-sign form

The next identity compares determinants of different orders. It is the main tool in the final exercise.

Problem. Let $X \in M_{n,k}(\mathbb{R})$ and $Y \in M_{k,n}(\mathbb{R})$. Prove

$$\det(I_n + XY) = \det(I_k + YX).$$

Solution. Consider

$$M = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}.$$

Eliminating the lower-left block gives

$$\begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix} M = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}.$$

The eliminating factor has determinant 1, so

$$\det M = \det(I_k + YX).$$

Alternatively, eliminate the upper-right block:

$$\begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix} M = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

This eliminating factor also has determinant 1, giving

$$\det M = \det(I_n + XY).$$

Comparing the two expressions proves the desired identity. Replacing Y by $-Y$ gives the minus-sign version in the textbook:

$$\det(I_n - XY) = \det(I_k - YX).$$

Dimensions and hypotheses

Although X and Y need not be square, the products XY and YX are square, of orders n and k, respectively. Neither $I_n + XY$ nor $I_k + YX$ is assumed invertible. The identity therefore applies at singular parameter values as well.

When k is small, it converts an $n \times n$ determinant into a much smaller one. For example, with $k = 1$, a column $u \in \mathbb{R}^n$ and a row v^T give

$$\det(I_n + uv^T) = \det(1 + v^T u) = 1 + v^T u.$$

The final exercise uses two such outer products simultaneously, reducing the computation to a 2×2 determinant.

9. A structured fourth-order determinant

Exercise 3.3, No. 13(1)

Problem. Use a matrix square to compute $\det M$, where

$$M = \begin{pmatrix} a & b & c & d \\ -b & a & d & -c \\ -c & -d & a & b \\ -d & c & -b & a \end{pmatrix}, \qquad a, b, c, d \in \mathbb{R}.$$

Solution. Separate the diagonal scalar part from the skew-symmetric part:

$$M = aI_4 + K, \qquad K = \begin{pmatrix} 0 & b & c & d \\ -b & 0 & d & -c \\ -c & -d & 0 & b \\ -d & c & -b & 0 \end{pmatrix}.$$

Then $K^T = -K$. Put $r^2 = b^2 + c^2 + d^2$. Direct multiplication yields

$$K^2 = -r^2 I_4.$$

For instance, $(K^2)_{11} = -b^2 - c^2 - d^2$, and the first-row off-diagonal entries vanish because

$$-cd + dc = 0, \qquad bd - db = 0, \qquad -bc + cb = 0.$$

The other entries have the same cancellations. Therefore

$$\begin{aligned} M^T M &= (aI_4 - K)(aI_4 + K) \\ &= a^2 I_4 - K^2 = (a^2 + b^2 + c^2 + d^2) I_4. \end{aligned}$$

Taking determinants gives

$$(\det M)^2 = (a^2 + b^2 + c^2 + d^2)^4.$$

This determines the absolute value of $\det M$; the sign still needs to be fixed.

Fix b, c, d and define the real polynomials

$$P(t) = \det(tI_4 + K), \qquad Q(t) = (t^2 + r^2)^2.$$

The same calculation for every real t gives $P(t)^2 = Q(t)^2$, hence

$$(P - Q)(P + Q) = 0$$

as a polynomial identity. Both P and Q have leading term t^4, so $P + Q$ is nonzero. A product of two nonzero real polynomials cannot be zero; thus $P = Q$. Substituting $t = a$ proves

$$\det M = (a^2 + b^2 + c^2 + d^2)^2.$$

The square was applied to $K = M - aI_4$. In general M^2 is not a scalar matrix; rather, $M^2 = 2aM - (a^2 + b^2 + c^2 + d^2) I_4$.

10. A determinant with two outer products

Exercise 3.3, No. 12(2)

Problem. For arbitrary real numbers $a_1, \dots, a_n, b_1, \dots, b_n$, evaluate

$$D = \begin{vmatrix} 1 + a_1 + b_1 & a_1 + b_2 & \cdots & a_1 + b_n \\ a_2 + b_1 & 1 + a_2 + b_2 & \cdots & a_2 + b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n + b_1 & a_n + b_2 & \cdots & 1 + a_n + b_n \end{vmatrix}.$$

Solution. Define the column vectors

$$a = (a_1, \dots, a_n)^T, \qquad b = (b_1, \dots, b_n)^T, \qquad \mathbf{1} = (1, \dots, 1)^T.$$

The entry in position (i, j) is $\delta_{ij} + a_i + b_j$, so the matrix whose determinant is required is

$$H = I_n + a\mathbf{1}^T + \mathbf{1}b^T.$$

The part beyond the identity is a sum of two outer products. Factor it as

$$H = I_n + UV, \qquad U = (a\ \mathbf{1}) \in M_{n,2}(\mathbb{R}), \qquad V = \begin{pmatrix} \mathbf{1}^T \\ b^T \end{pmatrix} \in M_{2,n}(\mathbb{R}).$$

Multiplying in the reverse order produces only a 2×2 matrix:

$$VU = \begin{pmatrix} \mathbf{1}^T a & \mathbf{1}^T \mathbf{1} \\ b^T a & b^T \mathbf{1} \end{pmatrix} = \begin{pmatrix} \displaystyle\sum_i a_i & n \\ \displaystyle\sum_i a_i b_i & \displaystyle\sum_i b_i \end{pmatrix}.$$

By the identity proved in Section 8,

$$\begin{aligned} D &= \det(I_n + UV) = \det(I_2 + VU) \\ &= \begin{vmatrix} 1 + \displaystyle\sum_i a_i & n \\ \displaystyle\sum_i a_i b_i & 1 + \displaystyle\sum_i b_i \end{vmatrix}. \end{aligned}$$

Therefore

$$D = \left(1 + \sum_{i=1}^{n} a_i\right)\left(1 + \sum_{i=1}^{n} b_i\right) - n \sum_{i=1}^{n} a_i b_i.$$

No division was used, so the formula holds for every choice of the parameters. As checks, $n = 1$ gives $1 + a_1 + b_1$, and $b = 0$ gives the rank-one formula $\det(I_n + a\mathbf{1}^T) = 1 + \sum_i a_i$.

Reference. Exercise labels follow Nan-Hua Xi, *Linear Algebra*, §3.3, Nos. 7, 8, 13(1), and 12(2), in the order of the handwritten Note 5.